Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

TEST REPORT FORM

Earth Electrode Resistance Test

Fall-of-potential method · 62 % rule with 52 / 62 / 72 % plateau check

01 Job and test details

Customer / site

Electrode ID / tag

Test date

Electrode type and description

Largest dimension of electrode L (m)

Distance E to C spike: D (m)

Soil condition

☐ Dry ☐ Damp ☐ Wet

Last rainfall

Permitted earth resistance (Ω)

Seasonal factor (default 1.0)

Test instrument, serial no.

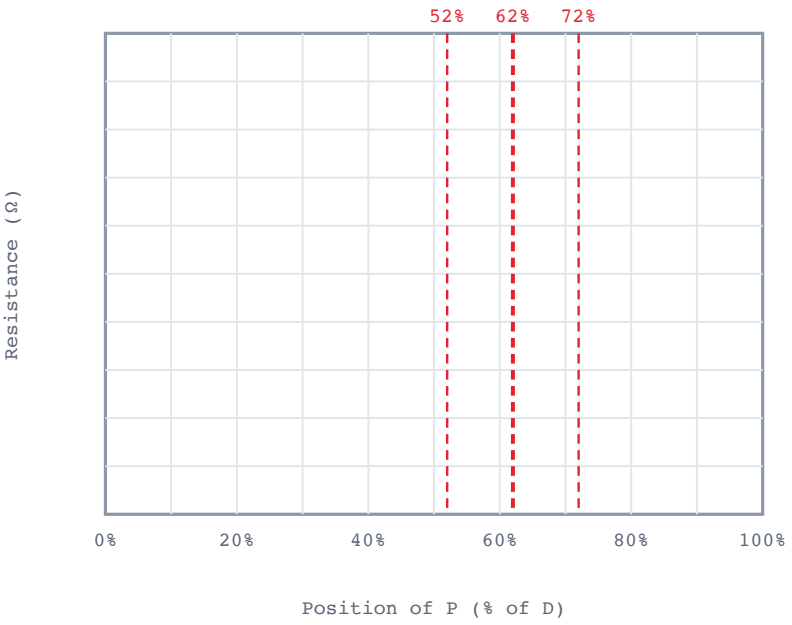
Tested by

Calibration due

02 Readings along the line from E to C

keep E and C fixed, move P in steps

P position % of D	P distance metres = % × D	Reading R Ω
10%		
20%		
30%		
40%		
50%		
52%		
62%		
72%		
80%		
90%		



The three shaded rows are the plateau check. Plot your readings on the graph: a flat region around 62 % means a good test.

03 Result

R at 52 %

R at 62 %

R at 72 %

Largest difference from R62

`max(|R52−R62|, |R72−R62|)/R62`

Plateau check (agree within 5 %) ☐ Valid ☐ Not valid

D at least 5 × L ☐ Yes ☐ No

Earth resistance = R62 × factor

Against permitted value ☐ PASS ☐ FAIL

Remarks and recommended action

Signature and date: tester

Signature and date: reviewer

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WORKSHEET AND REFERENCE

Soil resistivity and rod estimate

Wenner four-pin method, depth correction and a single-rod resistance estimate for comparison

01 Wenner soil resistivity survey

Traverse direction and location

Probe depth b (m)

Test date

Spacing a metres	Reading R Ω	b / a b divided by a	Simple resistivity 2π a R	Factor F from table	Corrected resistivity simple × F, Ω·m

Average ρ corrected (Ω·m)

Lowest and highest (Ω·m)

02 Depth correction factor F

F depends only on b / a

b / a	0.01	0.02	0.05	0.10	0.15	0.20	0.30
Factor F	1.0002	1.0007	1.0044	1.0172	1.0380	1.0659	1.1382

Exact Wenner relation: ρ = 4πaR / (1 + 2a / √(a² + 4b²) - a / √(a² + b²)), which equals 2πaR × F with F = 2 / (1 + 2 / √(1 + 4(b/a)²) - 1 / √(1 + (b/a)²)). For b much smaller than a, F is close to 1. Apparent resistivity at spacing a reflects soil to a depth of the order of a. A strong change with spacing suggests layered soil.

03 Single vertical rod estimate

for comparison with your measurement

Resistivity ρ (Ω·m)

Rod length in soil L (m)

Rod radius r (m)
`diameter / 2`

Estimated R (Ω)

Dwight formula
`R = ϱ / (2πL) × (ln(4L / r) − 1)`

Worked example (made-up numbers)
`ϱ = 60, L = 3 m, 16 mm rod: R = 20.1 Ω`

An estimate for one vertical rod in uniform soil. A difference of tens of percent from the measurement is common. A large difference points to layered soil, rock, moisture or interference worth investigating.

04 Method notes

- Keep the P and C spikes in line with the electrode, away from buried pipes and cables, and in undisturbed soil. If the three readings at 52, 62 and 72 % do not agree, increase D and repeat.
- Earth resistance changes with season and soil moisture. Record the soil condition and last rainfall, and set the seasonal factor from your site or utility practice.
- Limits and constants: IEEE 81-2012 (fall-of-potential and Wenner methods) and the Dwight rod formula. The 5 % plateau tolerance and 5 × L spacing are practical guides, not limits from a standard.

CIE INSTRUMENTS FOR THIS TEST

DET-2000
Digital earth resistance tester, 4-terminal, rechargeable battery

CIE/222M
Analog hand-driven earth tester, 3-terminal and 4-terminal (soil resistivity)

Check it online
Scan for the free soil resistivity calculator.
cieinstruments.in/tools